

SDK MANUAL

Neeuro OS SDK for Windows

Version S2.0.0-RC7

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(A) About

The NEEURO SenzeBand SDK is for developers who want to use the NEEURO SenzeBand to create apps or games that incorporate EEG technology.

Neeuro SenzeBand can be connected by using Bluetooth Low Energy (BLE) connection.

In Neeuro SenzeBand SDK Standard version, the SDK will analyse the data received and provide the following output to be used in the application.

1. Mental State - Attention classification
2. Mental State - Relaxation classification
3. Mental State - Workload classification
4. Raw EEG data (250 data in 1sec)
5. Raw EEG data 200ms (50 data in 200ms)
6. Filtered EEG data (with 1-40Hz Bandpass filter & 50Hz Notch filter)
7. Accelerometer values - X, Y, Z axes
8. Normalised Frequency waves (Delta, Theta, Alpha, Beta, Gamma, LowAlpha, HighAlpha, LowBeta, HighBeta)
9. Raw Frequency waves (Delta, Theta, Alpha, Beta, Gamma, LowAlpha, HighAlpha, LowBeta, HighBeta)
10. MCUID - A unique 32 character identifier string for the SenzeBand device
11. Battery level
12. Channel signal status - (T/F) strong EEG signal detected on 4 channels
13. Signal ready status - (T/F) sufficient strong EEG signal received for signal processing for Mental States
14. Connection status - (T/F) Good connection on BLE, no data lost
15. Orientation values - X, Y, Z axes*
16. Oxygen level and Heart rate*
17. Raw PPG data (IR and Red data)*

*Additional functions for SenzeBand 2

Please enter the provided Developer Code in the App.config file. The AuthenticationResult delegate will receive a True value when authentication succeeds. Please also ensure your computer has an internet connection.

(B) System Requirements

Minimum OS :
Windows 10

Developed on:
Visual Studio 2022

(C) SenzeBand Features and Use Info

Neeuro OS SenzeBand Software Development Kit (SDK)

The SenzeBand's sensors has a sampling rate of 250Hz, ie. each sensor produces 250 sampling values in 1 second.

The SenzeBand's Bluetooth transfer rate with the smart device is 1Hz, ie. the SenzeBand sends 1 data packet to the smart device every second.

The data packet contains raw data for 4 sensors, hence 4 sets of 250 values. Each set holds the signal values at every 1/250 seconds.

The accelerometer values also come at 1 Hz, ie. 1 sample in a data packet every second.

The frequency wave are:

Frequency wave	Frequency
Delta	1 – 4 Hz
Theta	4 – 8 Hz
Alpha	8 – 12 Hz
- Low Alpha	- 8 – 10 Hz
- High Alpha	- 10 – 12 Hz
Beta	12 – 30 Hz
- Low Beta	- 12 – 20 Hz
- High Beta	- 20 – 30 Hz
Gamma	30 – 40 Hz

(D) Setup for Project

- 1) Reference the dll: SenzeBandWindowsIPC_ClientDLL
- 2) Ensure that win_res is in the executable folder. It is required for Senzeband communication.

(F) Interface

View html documentation for more info.

(G) Support contact

For support, please email to support@neeuro.com